OLIST ANALYSIS

1. **TOTAL CUSTOMERS**

```
select count(distinct customer_unique_id) as total_customers
from olist_customers_dataset
```

total_customers
96096

2. **TOTAL ORDERS**

```
select count(distinct order_id) as total_orders
from olist_orders_dataset;
```

total_orders
99441

3. **TOTAL PRODUCTS**

```
select count(distinct product_id) as total_products
from olist_products_dataset;
```

total_products
32951

4. **-- TOTAL SELLERS**

```
select count(distinct seller_id) as total_sellers
from olist_sellers_dataset;
```

total_sellers
3095

5. **TOTAL ITEMS SOLD**

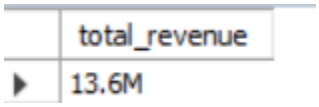
```
select count(order_item_id) as total_item_sold
from olist_order_items_dataset;
```

total_item_sold
112650

6. **TOTAL REVENUE**

```
select concat(round(sum(price)/1000000,1),"M") as total_revenue
```

from olist_order_items_dataset



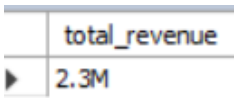
total_revenue
13.6M

7. TOTAL FREIGHT(Delivery cost)

select

concat(round(sum(freight_value)/1000000,1),"M") as total_freight

from olist_order_items_dataset;

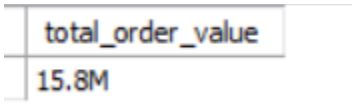


total_revenue
2.3M

8. TOTAL CUSTOMER SPEND (Total spent by customers)

select concat(round(sum(price+freight_value)/1000000,1),"M") as total_order_value

from olist_order_items_dataset;

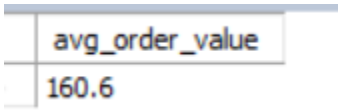


total_order_value
15.8M

9. AVG ORDER VALUE

select round(sum(price+freight_value)/count(distinct order_id),1) as avg_order_value

from olist_order_items_dataset;



avg_order_value
160.6

10. TOTAL ORDERS BY STATUS

select

order_status,

count(distinct order_id) total_orders

from

olist_orders_dataset

group by order_status

order by total_orders;

order_status	total_orders
approved	2
created	5
processing	301
invoiced	314
unavailable	609
canceled	625
shipped	1107
delivered	96478

11. REVENUE BY MONTH

```
select
round(sum(price),1) as total_revenue,
Year(order_purchase_timestamp) as `year`,
month(o.order_purchase_timestamp) as month_no,
monthname(o.order_purchase_timestamp) as month_name

from
olist_orders_dataset as o
left join olist_order_items_dataset as oi
on o.order_id = oi.order_id
group by
`year`,
month_no,
month_name
order by
`year`,
month_no,
month_name;
```

total_revenue	year	month_no	month_name
267.4	2016	9	September
49507.7	2016	10	October
10.9	2016	12	December
120312.9	2017	1	January
247303	2017	2	February
374344.3	2017	3	March
359927.2	2017	4	April
506071.1	2017	5	May
433038.6	2017	6	June
498031.5	2017	7	July
573971.7	2017	8	August
624401.7	2017	9	September
664219.4	2017	10	October
1010271.4	2017	11	November
743914.2	2017	12	December
950030.4	2018	1	January
844178.7	2018	2	February
983213.4	2018	3	March
996647.8	2018	4	April
996517.7	2018	5	May
865124.3	2018	6	June
895507.2	2018	7	July
854686.3	2018	8	August
145	2018	9	September
NULL	2018	10	October

12. ORDERS BY MONTH

```
select
count(distinct order_id) as total_orders,
Year(order_purchase_timestamp) as `year`,
month(order_purchase_timestamp) as month_no,
monthname(order_purchase_timestamp) as month_name

from olist_orders_dataset
group by
`year`,
month_no,
month_name
order by
`year`,
month_no
```

total_orders	year	month_no	month_name
4	2016	9	September
324	2016	10	October
1	2016	12	December
800	2017	1	January
1780	2017	2	February
2682	2017	3	March
2404	2017	4	April
3700	2017	5	May
3245	2017	6	June
4026	2017	7	July
4331	2017	8	August
4285	2017	9	September
4631	2017	10	October
7544	2017	11	November
5673	2017	12	December
7269	2018	1	January
6728	2018	2	February
7211	2018	3	March
6939	2018	4	April
6873	2018	5	May
6167	2018	6	June
6292	2018	7	July
6512	2018	8	August
16	2018	9	September
4	2018	10	October

13. REVENUE BY PRODUCT CATEGORY

```
select

round(sum(price),1) as total_revenue,

coalesce(pc.product_category_name_english,p.product_category_name, 'Unknown') as
product_category

from

olist_order_items_dataset as oi

left join olist_products_dataset as p

on oi.product_id = p.product_id

left join product_category_name_translation as pc

on p.product_category_name = pc.product_category_name

group by

product_category

order by

total_revenue desc;
```



total_revenue	product_category
1258681.3	health_beauty
1205005.7	watches_gifts
1036988.7	bed_bath_table
988049	sports_leisure
911954.3	computers_accessories
729762.5	furniture_decor
635290.8	cool_stuff
632248.7	housewares
592720.1	auto
485256.5	garden_tools
483946.6	toys
411764.9	baby
399124.9	perfumery
323667.5	telephony
273960.7	office_furniture
230943.2	stationery
222963.1	computers
214315.4	pet_shop
191498.9	musical_instruments
190648.6	small_appliances
179535.3	
160246.7	electronics
157465.2	consoles_games
152823.5	fashion_bags_accesso...
144677.6	construction_tools_co...

14. TOP 10 PRODUCTS CATEGORY BY REVENUE

```
select
row_number() over(order by sum(oi.price) desc) as num,
round(sum(oi.price),1) as total_revenue,
```

```
coalesce(pc.product_category_name_english,p.product_category_name, 'Unknown') as
product_category

from

olist_order_items_dataset as oi

left join olist_products_dataset as p

on oi.product_id = p.product_id

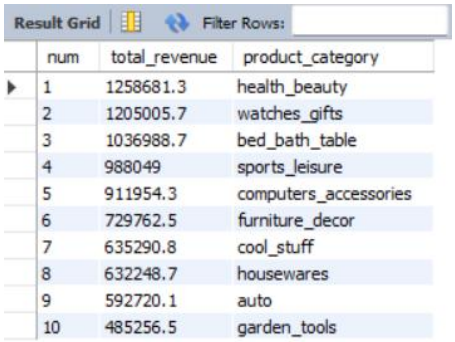
left join product_category_name_translation as pc

on p.product_category_name = pc.product_category_name

group by

product_category

limit 10;
```



Result Grid | Filter Rows:

num	total_revenue	product_category
1	1258681.3	health_beauty
2	1205005.7	watches_gifts
3	1036988.7	bed_bath_table
4	988049	sports_leisure
5	911954.3	computers_accessories
6	729762.5	furniture_decor
7	635290.8	cool_stuff
8	632248.7	housewares
9	592720.1	auto
10	485256.5	garden_tools

15. TOP 10 SELLER CITY BY REVENUE

```
select

round(sum(price),1) as total_revenue,

s.seller_city

from olist_sellers_dataset as s

left join olist_order_items_dataset as oi

on s.seller_id = oi.seller_id

group by s.seller_city

order by total_revenue desc

limit 10;
```



total_revenue	seller_city
2702925.9	sao paulo
624592.9	ibitinga
470759.8	curitiba
358413.6	rio de janeiro
329494.4	guarulhos
275976.4	ribeirao preto
230568.1	itaquaquecetuba
229472.6	guariba
228561.6	santo andre
225525	lauro de freitas

16. SALES BY CUSTOMER STATE

```
select
c.customer_state,
count(distinct o.order_id) as orders,
round(sum(oi.price),1) as revenue

from
olist_customers_dataset as c
left join olist_orders_dataset as o
on c.customer_id = o.customer_id
left join olist_order_items_dataset as oi
on o.order_id = oi.order_id
group by c.customer_state
order by revenue desc;
```

customer_state	orders	revenue
RS	5466	750304
PR	5045	683083.8
SC	3637	520553.3
BA	3380	511350
DF	2140	302603.9
GO	2020	294591.9
ES	2033	275037.3
PE	1652	262788
CE	1336	227254.7
PA	975	178947.8
MT	907	156453.5
MA	747	119648.2
MS	715	116812.6
PB	536	115268.1
PI	495	86914.1
RN	485	83035
AL	413	80314.8
SE	350	58920.9
TO	280	49621.7
RO	253	46140.6
AM	148	22356.8
AC	81	15982.9
AP	68	13474.3
RR	46	7829.4

17. PAYMENT METHOD ANALYSIS

```
select
payment_type,
count(distinct order_id) as transactions,
round(sum(payment_value),1) as payment_value
from olist_order_payments_dataset
group by payment_type
order by payment_value desc
```

payment_type	transactions	payment_value
credit_card	76505	12542084.2
boleto	19784	2869361.3
voucher	3866	379436.9
debit_card	1528	217989.8
not_defined	3	0

Result 1

18. PAYMENT INSTALLMENTS ANALYSIS

```
SELECT
```

payment_installments,

count(order_id) as transactions,

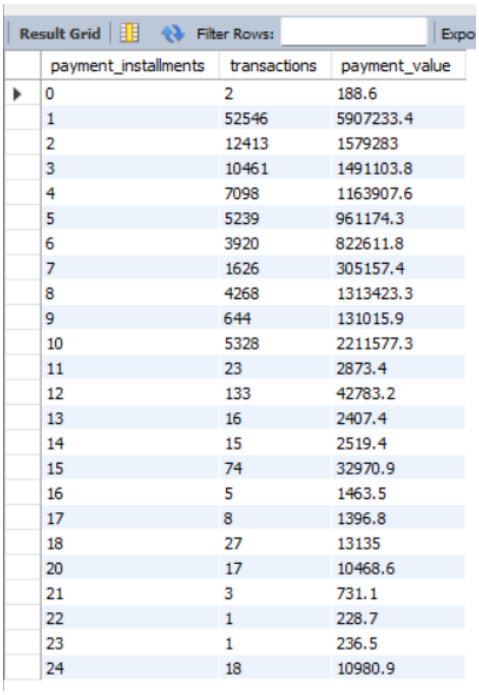
round(sum(payment_value),1) as payment_value

from olist_order_payments_dataset

group by payment_installments

payment_installments,

payment_value desc ;



Result Grid | Filter Rows: | Expo

payment_installments	transactions	payment_value
0	2	188.6
1	52546	5907233.4
2	12413	1579283
3	10461	1491103.8
4	7098	1163907.6
5	5239	961174.3
6	3920	822611.8
7	1626	305157.4
8	4268	1313423.3
9	644	131015.9
10	5328	2211577.3
11	23	2873.4
12	133	42783.2
13	16	2407.4
14	15	2519.4
15	74	32970.9
16	5	1463.5
17	8	1396.8
18	27	13135
20	17	10468.6
21	3	731.1
22	1	228.7
23	1	236.5
24	18	10980.9

19. AVG APPROVAL TIME

call all_olist_tables();

select

round(avg(timestampdiff(minute, order_purchase_timestamp, order_approved_at)/60

),

2)as avg_approval_hours

from olist_orders_dataset

where order_approved_at is not null;

avg_approval_hours
10.41

20. AVG PROCESSING TIME

```
select
round(avg(timestampdiff(hour, order_approved_at, order_delivered_carrier_date)/60
   ), 1)as avg_processing_minute
from olist_orders_dataset
where  order_approved_at is not null
and
order_delivered_carrier_date is not null;
```

avg_processing_hours
67.3

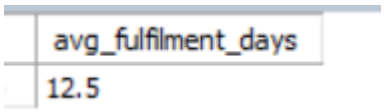
21. AVERAGE DELIVERY TIME

```
 select
round(avg(timestampdiff(hour,
order_delivered_carrier_date,order_delivered_customer_date)/24
   ),
   1)as avg_delivery_days
from olist_orders_dataset
where order_delivered_carrier_date is not null
and
order_delivered_customer_date is not null;
```

avg_delivery_days
9.3

22. AVG FULFILMENT TIME

```
select
round(avg(timestampdiff(hour, order_purchase_timestamp,
order_delivered_customer_date)/24
   ),
   1)as avg_fulfilment_days
from olist_orders_dataset
where order_purchase_timestamp is not null
and
order_delivered_customer_date is not null;
```



avg_fulfilment_days
12.5

23. DELIVERY TIME

```
call all_olist_tables();
select
case when
order_estimated_delivery_date >= order_delivered_customer_date
then "Ontime"
when order_delivered_customer_date > order_estimated_delivery_date
then "Late"
else "Unknown"
End as delivery_status,
count(distinct order_id) as orders

from
olist_orders_dataset
```

```
where order_estimated_delivery_date is not null
and
order_delivered_customer_date is not null
group by delivery_status
order by
orders desc ;
```

delivery_status	orders
Ontime	88649
Late	7827

24. ONTIME DELIVERY PERCENTAGE

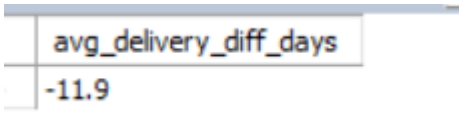
```
select
round(
100* sum(
case when order_estimated_delivery_date >= order_delivered_customer_date
then 1
else 0
End)/
count(distinct order_id),
1) as ontime_delivery_percentage
from
olist_orders_dataset
where order_estimated_delivery_date is not null
and
order_delivered_customer_date is not null
```



ontime_delivery_percentage
91.9

25. DELIVERY VARIANCE DAYS

```
select
round(avg(datediff(order_delivered_customer_date,order_estimated_delivery_date)),1
)as avg_delivery_variance_days
from olist_orders_dataset;
```



avg_delivery_diff_days
-11.9

26. REPEAT CUSTOMERS

```
select count(total_orders) AS repeat_customers

from
(
select
c.customer_unique_id,
count(distinct o.order_id) as total_orders
from olist_customers_dataset as c
left join
olist_orders_dataset as o
on c.customer_id = o.customer_id
group by c.customer_unique_id
having total_orders > 1
order by total_orders desc) t
```

;

repeat_customers
2997

27. REPEAT CUSTOMERS RATE

```
select count(total_orders)/
(
select count(distinct customer_unique_id) from olist_customers_dataset
)*100 as repeat_customer_rate

from
(
select
c.customer_unique_id,
count(distinct o.order_id) as total_orders
from olist_customers_dataset as c
left join
olist_orders_dataset as o
on c.customer_id = o.customer_id
group by c.customer_unique_id
having total_orders > 1
order by total_orders desc) t
```

repeat_customer_rate
3.1188

28. AVG REVIEW

```
call all_olist_tables();


select
avg(review_score) as avg_review
from olist_order_reviews_dataset
where review_score is not null
```

avg_review
4.0864

29. REVIEWS BY SCORE

```
select
review_score,
count(review_score) as review_score
from olist_order_reviews_dataset
group by review_score
order by review_score desc
```

review_score	review_score
5	57328
4	19142
1	11424
3	8179
2	3151

30. REVIEWS BY CATEGORY

```
call all_olist_tables();


select
coalesce(pc.product_category_name_english, nullif(p.product_category_name,''),
"unknown") as category,
```

```
avg(r.review_score) as avg_review
from
olist_order_reviews_dataset as r
left join olist_order_items_dataset as oi
on r.order_id = oi.order_id
left join olist_products_dataset as p
on oi.product_id = p.product_id
left join product_category_name_translation as pc
on p.product_category_name = pc.product_category_name
group by category
order by avg_review desc;
```

category	avg_review
cds_dvds_musicals	4.6429
fashion_childrens_clothes	4.5000
books_general_interest	4.4463
costruction_tools_tools	4.4444
flowers	4.4194
books_imported	4.4000
books_technical	4.3684
food_drink	4.3154
luggage_accessories	4.3153
small_appliances_home_oven_and_coffee	4.3026
fashion_sport	4.2581
fashion_shoes	4.2337
food	4.2182
music	4.2105
cine_photo	4.2055
stationery	4.1939
pet_shop	4.1851
computers	4.1750
home_appliances	4.1725
perfumery	4.1619
toys	4.1586
musical_instruments	4.1526
small_appliances	4.1492
cool_stuff	4.1463
fashion_bags_accessories	4.1447
home_appliances_2	4.1429
health_beauty	4.1428
arts_and_craftmanship	4.1250
tablets_printing_image	4.1235
furniture_bedroom	4.1182
sports_leisure	4.1080
industry_commerce_and_business	4.1015
signaling_and_security	4.0863
dvds_blu_ray	4.0794
auto	4.0655
housewares	4.0550

category	avg_review
housewares	4.0550
costruction_tools_garden	4.0542
construction_tools_lights	4.0541
construction_tools_construction	4.0529
drinks	4.0477
garden_tools	4.0427
electronics	4.0375
consoles_games	4.0231
christmas_supplies	4.0205
market_place	4.0194
watches_gifts	4.0192
baby	4.0118
la_cuisine	4.0000
agro_industry_and_commerce	4.0000
fashion_underwear_beach	3.9769
air_conditioning	3.9692
kitchen_dining_laundry_garden_furniture	3.9643
telephony	3.9469
home_construction	3.9400
art	3.9372
computers_accessories	3.9308
furniture_living_room	3.9044
furniture_decor	3.9035
bed_bath_table	3.8957
construction_tools_safety	3.8446
home_confort	3.8299
audio	3.8255
furniture_mattress_and_upholstery	3.8158
fashio_female_clothing	3.7800
party_supplies	3.7674
fixed_telephony	3.6832
fashion_male_clothing	3.6412
home_comfort_2	3.6296
office_furniture	3.4932
pc_gamer	3.3333
portateis_cozinha_e_preparadores_de_a...	3.2667
diapers_and_hygiene	3.2564

esult 77 ×

diapers_and_hygiene	3.2564
unknown	3.1599
security_and_services	2.5000

esult 77 ×